In Silico Closed-Loop Metabolic Gating of Glycolytic-to-Oxidative Transition in *Ex Vivo* CAR-T Perfusion Bioreactors Guided by Structural Bioinformatics and Optical Redox Observability

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Code & Reproducibility Repository: https://github.com/deesatzed/mitorestorer

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Abstract

Durable clinical responses in adoptive chimeric antigen receptor (CAR) T-cell therapy correlate with infusing stem-like memory T cells (T_{SCM}, $CD45RA^+CD62L^+CCR7^+TCF7^+$) possessing high mitochondrial spare respiratory capacity (SRC), yet 168-hour *ex vivo* perfusion expansion under continuous high glucose and interleukin-2 (IL-2) enforces a Warburg glycolytic program that accumulates depolarized mitochondria and TOX^+ epigenetic exhaustion. Here we present a strictly *in silico* systems-and-control investigation—combining structural bioinformatics, eight-state nonlinear ordinary differential equation (ODE) modeling, discrete-adjoint optimal control, physical inline sensor observability analysis, and freestanding fixed-point firmware verification—to design falsifiable closed-loop metabolic gating protocols prior to wet-lab execution. Heavy-atom analysis of experimental crystal structures confirms druggable allosteric pockets on pyruvate dehydrogenase kinase 1 (`PDK1`, `PDB: 2Q8G`, 1.90 Å; lipoyl-pocket `Ser75` hydrogen bond at 2.50 Å to `AZD7545`) and sirtuin 1 (`SIRT1`, `PDB: 4ZZJ`, 2.74 Å; `Glu230` allosteric hinge at 3.47 Å), whereas full-length AlphaFold v6 models of downstream effectors `FOXO1` (655 AA, mean pLDDT = 50.96) and `PGC-1α` (798 AA, mean pLDDT = 52.74) exhibit single structured nucleic-acid-binding modules flanked by intrinsically disordered post-translational modification regions (9/9 `Akt/SIRT1` sites on `FOXO1` and 39/47 lysines on `PGC-1α` have pLDDT < 70). Building on Paper I, our simulated two-stage **Bioreactor-BCR** controller separates Stage-1 mitophagy-assisted glycolytic blastogenesis (u_1) from Stage-2 pyruvate dehydrogenase reactivation and NAD^+/`SIRT1`-coupled oxidative consolidation (u_2), yielding 62.80× fold expansion, 73.20% T_{SCM}, SRC = 4.87×, and $h(168\text{ h}) = 6.73\%$ on the nominal model, closely tracking the 200-variable discrete-adjoint Pontryagin Pareto frontier (all-200-variable gradient error 2.04×10^{-10}, 5-start cost spread 1.12%). Across $N = 1{,}000$ simulated heterogeneous oncology apheresis donors with unmodeled kinetic dispersion and ±6% optical-density (`OD`) lymphoblast-size calibration bias, physical 4-channel sensing (`Lac:Pyr`, volumetric OUR_{vol}, `OD`, and pH) fails (40.5% 5-criteria pass rate) because cell-density bias and Warburg-rate heterogeneity confound specific respiration. Augmenting the observer with a ratiometric autofluorescence optical redox ratio (`ORR`)—in which cell concentration cancels analytically—restores state observability ($\hat{h}$ RMSE 2.08%), lifting the 168-hour pass rate from 54.1% (best fixed 68 h timer) to 58.7% (paired McNemar $p = 8.04 \times 10^{-4}$; vs. 4-channel $p = 1.19 \times 10^{-41}$), retaining 59.4% under 100% enzymatic `Lac:Pyr` biofouling via NIS > 13.5 optical handover ($p = 5.22 \times 10^{-129}$ vs. 12.3% without `ORR`), and reaching 74.0% ($p = 2.08 \times 10^{-33}$) when paired with `OD`-triggered adaptive harvest (168–240 h) for hypoproliferative donors. Finally, freestanding FPU-free C99 firmware implementing both the `Q16.16/Q8.24` $\Sigma\Delta$ `RK4` plant (0.0078%–0.0114% error) and the 5-channel sequential Huber-EKF (0.0216% parity vs. `FP64`) executes within a zero-heap 2,704 B (33.01%) static SRAM workspace and rescues simulated perfusion-line acidosis (min pH 6.14 → 6.91).

1 Introduction

1.1 Bioenergetic Determinants of CAR-T Persistence

In autologous CD19- and BCMA-directed chimeric antigen receptor (CAR) T-cell manufacturing, clinical durability in chronic lymphocytic leukemia, B-cell acute lymphoblastic leukemia, and non-Hodgkin lymphoma is governed less by total infused cell count than by the proportion of early stem-cell memory T lymphocytes (T_{SCM}, $CD8^+CD45RA^+CD62L^+CCR7^+TCF7^+$) and their mitochondrial spare respiratory capacity (SRC) [1, 2]. Transcriptomic and epigenetic studies have established two

converging mechanistic axes underlying this phenotype: first, nuclear retention of Forkhead box protein O1 (FOXO1) acts as an essential master regulator of CAR-T memory programming and metabolic fitness while opposing TOX^+ terminal exhaustion [3, 4, 5]; second, selective clearance of depolarized, fragmented mitochondria via PINK1/Parkin-dependent mitophagy (stimulated *ex vivo* or *in vivo* by urolithin A) preserves fused cristae networks and expands T_{SCM} subsets [6, 7, 8].

1.2 The Ex Vivo Proliferation–Exhaustion Conflict

Automated closed-cassette perfusion bioreactors (such as hollow-fiber or gas-permeable membrane cassettes) activate patient apheresis T cells via anti-CD3/CD28 stimulation over a 7-day (168 h) expansion cycle. This manufacturing window imposes a well-known dynamical tradeoff between clonal blastogenesis and bioenergetic stemness [9, 2]:

1. **Unphased Aerobic Glycolysis (Warburg Lock):** Continuous exposure to high glucose (25 mM) and high-dose IL-2 (300 IU/mL) drives mTORC1, Akt, and HIF-1α signaling. HIF-1α upregulates pyruvate dehydrogenase kinase 1 (PDK1) [10], which phosphorylates the pyruvate dehydrogenase (PDH) E1α subunit, shunting pyruvate to lactate ($\mathrm{Lac:Pyr} > 25$) and acidifying the perfusate. Concurrently, Akt phosphorylates FOXO1 at Thr24, Ser256, and Ser319, triggering 14-3-3 chaperone binding and nuclear export [3, 4], while mTORC1 suppresses ULK1-initiated mitophagy, allowing depolarized organelles (D) and TOX^+ chromatin remodeling to accumulate [9, 5].

2. **Premature Oxidative Forced Switching (Dose Failure):** Conversely, forcing oxidative phosphorylation or restricting glycolytic flux immediately upon CD3/CD28 activation ($t = 0$ h) deprives dividing lymphoblasts of glycolytic ribonucleotide, serine, and lipid precursors required during early log-phase expansion, risking failure to reach minimum clinical release doses ($\geq 50\times$ viable fold expansion).

3. **Oncology Apheresis Heterogeneity and Sensor Drift:** Unlike healthy-donor research runs, heavily pretreated oncology patients enter apheresis with wide baseline dispersion in T-cell mitochondrial depolarization ($h_0 \in [0.20, 0.62]$) and division kinetics [1]. Consequently, a fixed open-loop media-switch timer tuned to a nominal donor risks switching slow-clearing donors prematurely or holding rapid-proliferating donors in prolonged glycolysis. Yet closed-loop gating using inline enzymatic supernatant sensors faces its own bioprocess hazard: amperometric lactate/pyruvate electrodes undergo progressive protein biofouling over multi-day perfusion.

1.3 Epistemic Scope and Computational Contributions

Disclosure of Computational Scope: This study is strictly a theoretical, structural-bioinformatic, and *in silico* control-engineering investigation. *No wet-lab, in vitro, ex vivo, or in vivo biological experiments were performed.* All cell-expansion trajectories, heterogeneous patient-cohort statistics, biofouling fault responses, and embedded processor benchmarks are generated by deterministic mathematical models and host-emulated firmware designed to formulate quantitative, falsifiable predictions for subsequent experimental testing (§7).

Extending the nondimensional staged heteroplasmy control theory of Paper I [11] to proliferating cell populations coupled to extracellular bioreactor mass transfer, this paper makes four specific contributions:

1. **Unified 8-State Perfusion Cassette ODE with 1-to-1 Code Parity (§2):** We formulate an eight-state bioreactor model coupling intracellular mitochondrial subpopulations (W, D), clonal cell fold expansion (N_{cells}), supernatant lactate-to-pyruvate ratio (R_{LP}), nuclear FOXO1 fraction (F_{nuc}), cristae-linked SRC (S_{SRC}), epigenetic exhaustion (E_{TOX}), and perfusate pH, with every closed-form auxiliary equation verified identically across FP64 Python, smooth discrete-adjoint optimization, and bare-metal C99 fixed-point code.

2. **Atomic Contact & AlphaFold v6 Disorder Audit Guiding Actuator Selection (§3):** By auditing heavy-atom ligand contacts in PDB: 2Q8G (PDK1–AZD7545) and PDB: 4ZZJ (SIRT1–STAC–carba-NAD^+) alongside AlphaFold Database v6 pLDDT and Predicted Aligned Error (PAE) matrices for full-length FOXO1 and PGC-1α, we quantify why disordered regulatory PTM tails lack preformed small-molecule pockets and why upstream enzymatic actuation—paired with explicit controls for known STAC chemical-probe artifacts [12, 13]—provides a structurally grounded intervention strategy.

3. **Discrete-Adjoint Pareto Frontier, Ratiometric ORR Observability & Extended-Horizon Rescue (§4–§5):** Solving a 200-variable discrete-adjoint optimal control problem (L-BFGS-B, all-200-variable gradient error 2.04×10^{-10}, 5-start spread 1.12%) maps the proliferation–stemness Pareto frontier. Across $N = 1{,}000$ heterogeneous oncology donors under physical volumetric $\mathrm{OUR_{vol}}$ and $\pm 6\%$ OD scattering bias, we prove why cell concentration cancels out identically in ratiometric autofluorescence ORR [14], lifting the 168-hour pass rate from 40.5% (4-channel No-ORR) and 54.1% (best fixed 68 h timer) to 58.7% (exact McNemar $p = 8.04 \times 10^{-4}$), retaining 59.4% under 100% Lac:Pyr biofouling via NIS > 13.5 optical handover ($p = 5.22 \times 10^{-129}$ vs. 12.3% without

ORR), and reaching 74.0% ($p = 2.08 \times 10^{-33}$) under OD-guided adaptive harvest (168–240 h).

4. **Plant-Preserving Acidosis Flush & Deterministic FPU-Free C99 Safety Firmware (§6):** Eliminating unphysical state-reset assumptions, we verify a continuous-plant neutral-buffer bypass flush (u_{fl}) during simulated perfusion-line occlusion and demonstrate freestanding FPU-free C99 execution of both the Q16.16/Q8.24 ΣΔ RK4 integrator and the 5-channel sequential Huber-EKF (0.0216% parity vs. FP64) within a zero-heap 2,704 B static SRAM workspace.

2 Mathematical Model of the 168-Hour Perfusion Cassette

2.1 Eight-State Cassette Vector and Bounded Controls

We model an automated closed-cassette CAR-T perfusion culture over $t \in [0, T_{\max}]$ ($T_{\max} = 168.0$ h) via the state vector

$$\mathbf{x}(t) = \big[W,\, D,\, N_{\mathrm{cells}},\, R_{\mathrm{LP}},\, F_{\mathrm{nuc}},\, S_{\mathrm{SRC}},\, E_{\mathrm{TOX}},\, \mathrm{pH}\big]^{\top}, \tag{1}$$

where $W(t) \geq 0$ is normalized intact, fused-cristae wild-type mitochondrial mass per cell, $D(t) \geq 0$ is normalized depolarized/fragmented mitochondrial mass per cell, $N_{\mathrm{cells}}(t) \geq 1$ is viable clonal fold expansion ($N_{\mathrm{cells}}(0) = 1.0$), $R_{\mathrm{LP}}(t) > 0$ is the extracellular supernatant molar [Lactate]/[Pyruvate] ratio, $F_{\mathrm{nuc}}(t) \in [0, 1]$ is the nuclear-retained, transcriptionally active fraction of FOXO1, $S_{\mathrm{SRC}}(t) \geq 0$ is mitochondrial spare respiratory capacity (dimensionless multiple of basal oxygen consumption), $E_{\mathrm{TOX}}(t) \in [0, 1]$ is the fraction of cells committed to $\mathrm{TOX}^{+}\mathrm{PD\text{-}1}^{+}\mathrm{TIM\text{-}3}^{+}$ epigenetic exhaustion, and $\mathrm{pH}(t)$ is the cassette perfusate pH. Total normalized organelle mass is $M(t) = W(t) + D(t)$, with depolarized fraction (functional heteroplasmy) $h(t) = D(t)/M(t)$ and intact fraction $w_{\mathrm{frac}}(t) = 1 - h(t) = W(t)/M(t)$.

The cassette is actuated by three bounded inputs $\mathbf{u}(t) = [u_1(t), u_2(t), u_{\mathrm{fl}}(t)]^{\top} \in [0, 1]^3$:

- $u_1(t) \in [0, 1]$: **Stage-1 Mitophagy & Mild Glycolytic Restraint Perfusate (Media-A)**, modeling urolithin A (25 μM) mitophagy induction paired with moderate PI3Kδ/Akt restraint during high-glucose blastogenesis [7, 4];
- $u_2(t) \in [0, 1]$: **Stage-2 Oxidative Stemness Perfusate (Media-B)**, modeling iso-osmotic partial galactose substitution ($\alpha_{\mathrm{gal}} = 0.74$), allosteric PDK1 lipoyl-pocket inhibition (AZD7545), NAD^{+} repletion / SIRT1 activation, and homeostatic IL-7/IL-15 cytokine support [10, 15];
- $u_{\mathrm{fl}}(t) \in [0, 1]$: **Emergency Neutral-Buffer Bypass Flush Valve**, routing fresh $\mathrm{pH}_{\mathrm{media}} = 7.36$ buffer across the hollow-fiber cartridge if main perfusion-line occlusion triggers acute lactic acidosis.

2.2 Closed-Form Auxiliary Bioenergetic Equations

To guarantee zero discrepancy between analytical expressions, eval_cart8_rhs_vec in Python, and eval_cart8_rhs_q16_to_q24 in C99, every intermediate rate is specified explicitly:

$$g_{\mathrm{glc}}(u_2) = 1 - \alpha_{\mathrm{gal}}\, u_2, \quad g_{\mathrm{pdh}}(u_2) = 0.15 + 0.85\, u_2, \tag{2}$$

$$v_{\mathrm{unmask}}(R_{\mathrm{LP}}) = \frac{K_{\mathrm{LP}}}{K_{\mathrm{LP}} + (R_{\mathrm{LP}} - R_{\mathrm{basal}})_{+}}, \tag{3}$$

$$k_{\mathrm{mito},D}(u_1, R_{\mathrm{LP}}) = k_{\mathrm{mito},W} + (k_{\mathrm{mito},D,\max} - k_{\mathrm{mito},W})\, u_1\, v_{\mathrm{unmask}}(R_{\mathrm{LP}}), \tag{4}$$

$$\mathrm{ATP}_{\mathrm{ox}}(\mathbf{x}, u_2) = (W + \eta_D D)(1 + \gamma_{\mathrm{pdh}} u_2) \times \big(1 + \gamma_{\mathrm{src}}(S_{\mathrm{SRC}} - 1)_{+}\big), \tag{5}$$

$$\mathrm{ATP}_{\mathrm{glyc}}(\mathbf{x}, \mathbf{u}) = \kappa_{\mathrm{glyc}}\, g_{\mathrm{glc}}(u_2)\big(1 - c_{\mathrm{akt}} u_1 (1 - u_2)\big) \times \frac{R_{\mathrm{LP}}}{R_{\mathrm{LP}} + K_{R,\mathrm{ATP}}}, \tag{6}$$

$$\phi_{\mathrm{pH}}(\mathrm{pH}) = \frac{1}{1 + \big((7.02 - \mathrm{pH})_{+}/0.16\big)^{2}}, \tag{7}$$

$$\dot{e}_{\mathrm{load}}(\mathbf{x}) = \Big[1 + \beta_{\mathrm{LP}} \frac{(R_{\mathrm{LP}} - R_{\mathrm{basal}})_{+}}{K_{\mathrm{LP}}} + \beta_h h + \beta_{\mathrm{pH}}(7.02 - \mathrm{pH})_{+}\Big] \times \big[\max(0.26,\, W + \eta_D D)\big]^{-1}, \tag{8}$$

$$\mathrm{EP}(\mathbf{x}, \mathbf{u}) = \dot{e}_{\mathrm{load}}(\mathbf{x})\,(1 - c_1 u_1 + c_2 u_2)^2, \tag{9}$$

$$w_{\mathrm{func}}(\mathbf{x}) = W(1 - h),$$

$$\Phi_{\mathrm{ox}}(\mathbf{x}) = \frac{w_{\mathrm{func}}^2}{w_{\mathrm{func}}^2 + K_{W,\mathrm{div}}^2}. \tag{10}$$

Here $v_{\mathrm{unmask}}(R_{\mathrm{LP}})$ captures Warburg/mTORC1 suppression of ULK1/PINK1 mitophagic flux under excessive lactate:pyruvate accumulation, ϕ_{pH} captures severe proton-mediated inhibition of blastogenesis and nuclear import below pH 7.02, and Φ_{ox} requires intact cristae mass w_{func} to sustain rapid oxidative division in low-glucose Stage-2 media.

2.3 Coupled Governing ODEs and Readout Map

With total cellular ATP availability $\mathrm{ATP}_{\mathrm{tot}} = \mathrm{ATP}_{\mathrm{ox}} + \mathrm{ATP}_{\mathrm{glyc}}$ and damage conversion rate $d_{\mathrm{ros}}(\mathbf{x}, \mathbf{u}) =$

$k_{\mathrm{dmg}}\max(0,\mathrm{EP}-\theta_{\mathrm{GSH}})$, the eight-state ODE system is:

$$\dot{W} = \big(k_{\mathrm{bio},0} + (k_{\mathrm{bio,max}} - k_{\mathrm{bio},0})\,u_2 F_{\mathrm{nuc}}\big) W\Big(1-\frac{M}{K}\Big)_+ - \big(k_{\mathrm{mito},W} + d_{\mathrm{ros}}(\mathbf{x},\mathbf{u})\big)W, \tag{11}$$

$$\dot{D} = k_{\mathrm{rep,adv}}\big(k_{\mathrm{bio},0} + \xi_D(k_{\mathrm{bio,max}} - k_{\mathrm{bio},0})\,u_2\big) \times D\Big(1-\frac{M}{K}\Big)_+ - k_{\mathrm{mito},D}(u_1, R_{\mathrm{LP}})\,D + d_{\mathrm{ros}}(\mathbf{x},\mathbf{u})\,W, \tag{12}$$

$$\dot{N}_{\mathrm{cells}} = \Big[\mu_{\mathrm{glyc}} g_{\mathrm{glc}}(u_2)\big(1-0.08u_1(1-u_2)\big) + \mu_{\mathrm{ox}} u_2 g_{\mathrm{pdh}}(u_2)\Phi_{\mathrm{ox}}\Big] \times \phi_{\mathrm{pH}}\,\frac{\mathrm{ATP}_{\mathrm{tot}}}{\mathrm{ATP}_{\mathrm{tot}}+K_{\mathrm{ATP}}}\,N_{\mathrm{cells}}\Big(1-\frac{N_{\mathrm{cells}}}{N_{\mathrm{cap}}}\Big)_+, \tag{13}$$

$$\dot{R}_{\mathrm{LP}} = k_{\mathrm{Warburg}} g_{\mathrm{glc}}(u_2)\big(1-0.35u_1(1-u_2)\big)(1+\alpha_h h) - k_{\mathrm{PDH}}(1+\beta_{\mathrm{azd}}u_2)w_{\mathrm{frac}}R_{\mathrm{LP}} - (k_{\mathrm{perf,LP}} + k_{\mathrm{fl,LP}}u_{\mathrm{fl}})(R_{\mathrm{LP}} - R_{\mathrm{media}}), \tag{14}$$

$$\dot{F}_{\mathrm{nuc}} = k_{\mathrm{imp}}\big(f_0 + \alpha_{\mathrm{akt}}u_1 + \phi_{\mathrm{SIRT1}}u_2 g_{\mathrm{pdh}}(u_2)\big) \times \phi_{\mathrm{pH}}\,(1-F_{\mathrm{nuc}}) - k_{\mathrm{exp}}(1-0.52u_1)\big(1-0.85u_2 g_{\mathrm{pdh}}(u_2)\big) \times \big(1+\gamma_{\mathrm{acid}}(6.90-\mathrm{pH})_+\big)F_{\mathrm{nuc}}, \tag{15}$$

$$\dot{S}_{\mathrm{SRC}} = s_{\mathrm{basal}}(W+\eta_D D) - k_{\mathrm{src,dec}}\,S_{\mathrm{SRC}} + k_{\mathrm{cristae}} w_{\mathrm{func}}\big(1+\beta_{\mathrm{src}}u_2 F_{\mathrm{nuc}} g_{\mathrm{pdh}}(u_2)\big), \tag{16}$$

$$\dot{E}_{\mathrm{TOX}} = \Big[k_{\mathrm{exh}}(\mathrm{EP}-\theta_{\mathrm{exh}})_+ + k_{\mathrm{acid,exh}}(6.86-\mathrm{pH})_+ + k_{\mathrm{warb,exh}} g_{\mathrm{glc}}(u_2)(1-0.22u_1)\Big(1+0.70\frac{N_{\mathrm{cells}}}{N_{\mathrm{ref}}}\Big)\Big] \times (1-E_{\mathrm{TOX}}) - \big(r_{\mathrm{epig},0} + k_{\mathrm{rev,epig}}u_2 F_{\mathrm{nuc}} g_{\mathrm{pdh}}(u_2)\big)E_{\mathrm{TOX}}, \tag{17}$$

$$\dot{\mathrm{pH}} = (k_{\mathrm{perf},0} + k_{\mathrm{perf,fl}}u_{\mathrm{fl}})(\mathrm{pH}_{\mathrm{media}} - \mathrm{pH}) - k_{\mathrm{acid}} g_{\mathrm{glc}}(u_2)(R_{\mathrm{LP}} - R_{\mathrm{buf}})_+ \frac{N_{\mathrm{cells}}}{N_{\mathrm{ref}}}, \tag{18}$$

where $(z)_+ \equiv \max(0,z)$. The terminal stem-like memory CAR-T fraction (%) is evaluated via the algebraic phenotypic readout

$$\mathrm{T}_{\mathrm{SCM}}(\mathbf{x}) = 100\,F_{\mathrm{nuc}}\,(1-E_{\mathrm{TOX}})\,\frac{S_{\mathrm{SRC}}}{S_{\mathrm{SRC}}+K_{\mathrm{SRC,mem}}}, \tag{19}$$

reflecting the joint requirement of nuclear FOXO1 retention (F_{nuc}), absence of TOX$^+$ epigenetic scarring ($1-E_{\mathrm{TOX}}$), and bioenergetic cristae reserve (S_{SRC}) [3, 4, 2].

Proposition 1 (Stage-1 Quotient Selection) *During Stage-1 mitophagy-primed expansion ($u_1 = 1, u_2 = 0$) whenever $\mathrm{EP}(\mathbf{x},\mathbf{u}) \le \theta_{\mathrm{GSH}}$ so $d_{\mathrm{ros}} = 0$, differentiating $h = D/(W+D)$ along (11)–(12) yields the exact Riccati quotient law of Paper I [11]:*

$$\dot{h}(t) = -\lambda_{\mathrm{CART}}(t)\,h(t)\big(1-h(t)\big), \tag{20}$$

with time-varying selective clearance rate

$$\lambda_{\mathrm{CART}}(t) = \big(k_{\mathrm{mito},D}(1, R_{\mathrm{LP}}(t)) - k_{\mathrm{mito},W}\big) - (k_{\mathrm{rep,adv}}-1)k_{\mathrm{bio},0}\Big(1-\frac{M(t)}{K}\Big)_+. \tag{21}$$

Consequently, in terms of the accumulated-selection coordinate $s(t) = \int_0^t \lambda_{\mathrm{CART}}(q)\,dq$, intracellular depolarization evolves exactly as $h(s) = [1 + \frac{1-h_0}{h_0}e^s]^{-1}$, reaching any target $h_{\mathrm{gate}} \in (0, h_0)$ at $s^* = \ln\big(\frac{h_0(1-h_{\mathrm{gate}})}{h_{\mathrm{gate}}(1-h_0)}\big)$.

Notice in (15)–(17) why *co-actuation* of PDK1 inhibition (g_{pdh}) and SIRT1/NAD$^+$ activation is required in Stage 2: without PDK1 inhibition ($u_{\mathrm{azd}} = 0 \implies g_{\mathrm{pdh}} = 0.15$), pyruvate remains blocked from the TCA cycle, collapsing both oxidative proliferation $\mu_{\mathrm{ox}} g_{\mathrm{pdh}} \Phi_{\mathrm{ox}}$ and NAD$^+$-coupled FOXO1 nuclear retention; conversely, without SIRT1 activation ($u_{\mathrm{sirt}} = 0$), wild-type biogenesis remains at basal $k_{\mathrm{bio},0}$ while depolarized organelles expand with replicative advantage $k_{\mathrm{rep,adv}} = 1.22$ via $\xi_D u_2$, causing $h(t)$ to rebound during Stage 2. All nominal parameters, initial conditions $\mathbf{x}(0) = [0.54, 0.42, 1.00, 22.0, 0.28, 1.30, 0.08, 7.32]^\top$ ($h_0 = 43.75\%$), and explicit provenance tags—[**M**] mechanistic/biophysical literature scale, [**F**] calibrated to published T-cell dynamic range, [**A**] operational cassette hardware parameter—are documented in Table 1.

3 Structural Bioinformatics Guiding Actuator Selection

A central design question for Stage-2 metabolic consolidation is why pharmacological actuation must target upstream metabolic kinases and deacetylases (PDK1 and SIRT1/NAD$^+$) rather than attempting direct small-molecule pocket activation of the master stemness transcription factors FOXO1 and PGC-1α. To ground our actuator parameterization in atomic coordinates, we audited experimental macromolecular crystallographic information files (.cif) from the Protein Data Bank alongside full-length AlphaFold Database v6 models and $N \times N$ Predicted Aligned Error (PAE) matrices [17, 18] (Figure 1, Table 2).

3.1 Druggable Allosteric Pockets in PDK1 (2Q8G) and SIRT1 (4ZZJ)

Coordinate inspection of human PDK1 co-crystallized with the trifluoropropanamide inhibitor AZD7545 (PDB: 2Q8G, Chain A, 368 resolved residues spanning 41–423 at 1.90 Å resolution [10]) confirms that AZD7545 (AZX, 31 heavy atoms) occupies the hydrophobic N-terminal lipoyl-binding (L2) pocket rather than the C-terminal ATP-binding cleft. Twelve residues form heavy-atom contacts $\le$ 4.0 Å to AZD7545 (Figure 1A): a short 2.50 Å hydrogen bond between Ser75 (Oγ) and the

Table 1: **Complete Parameterization and Provenance of the 8-State *In Silico* CAR-T Perfusion Bioreactor Model.** Every value matches `NOMINAL_PARAMS` in `simulator/generate_paper2_verified_benchmarks.py` and `build/coralnpu/cart_bioreactor_hil_testbench.c` 1-to-1. Provenance tags: **[M]** primary literature scale; **[F]** calibrated phenomenological coupling; **[A]** operational bioreactor/cassette specification.

Subsystem	Parameter(s)	Nominal Value(s)	Units	Tag	Biophysical Role & Literature Basis
Mitochondrial (W, D)	K, $k_{\mathrm{mito},W}$, $k_{\mathrm{bio},0}$	1.40, 0.008, 0.028	norm., h^{-1}, h^{-1}	[M/F]	Organelle cap & basal turnover [2, 11]
Mitophagy & Biogen.	$k_{\mathrm{mito},D,\max}$, $k_{\mathrm{bio},\max}$, $k_{\mathrm{rep,adv}}$, ξ_D	0.068, 0.086, 1.22, 1.00	h^{-1}, h^{-1}, –, –	[F]	Urolithin A clearance & replicative bias [7, 16]
Warburg Masking	R_{basal}, K_{LP}, η_D	8.0, 22.0, 0.22	molar ratio, –, –	[F]	`mTORC1/Lac:Pyr` masking of `ULK1`; D OXPHOS efficiency
Energetic Stress	γ_{pdh}, γ_{src}, α_{gal}, κ_{glyc}, c_{akt}, $K_{R,\mathrm{ATP}}$	0.72, 0.22, 0.74, 1.40, 0.08, 8.0	dimensionless	[F]	OXPHOS/glycolytic ATP & galactose shift [9]
ROS Damage Gate	β_{LP}, β_h, β_{pH}, c_1, c_2, θ_{GSH}, k_{dmg}	0.40, 0.88, 1.90, 0.24, 0.06, 1.38, 0.0012	–, h^{-1}	[F]	Threshold ROS conversion $W \to D$ above θ_{GSH} [11]
Clonal Division N	μ_{glyc}, μ_{ox}, K_{ATP}, $K_{W,\mathrm{div}}$, N_{cap}	0.062, 0.0185, 0.52, 0.56, 85.0	h^{-1}, –, fold	[M/A]	Lymphoblast doubling ($T_d \approx 11$–15 h) & cassette cap [1]
Supernatant R_{LP}	k_{Warburg}, α_h, k_{PDH}, β_{azd}, R_{media}	3.00, 0.46, 0.088, 2.80, 5.0	h^{-1}, –, ratio	[M/F]	Aerobic glycolysis vs. `AZD7545` `PDH` flux [10]
`FOXO1` Shuttle F_{nuc}	k_{imp}, f_0, α_{akt}, ϕ_{SIRT1}, k_{exp}, γ_{acid}	0.056, 0.30, 0.30, 1.18, 0.032, 7.5	h^{-1}, –	[M/F]	`Akt` 14-3-3 export vs. `SIRT1` retention [3, 4]
Cristae SRC	s_{basal}, k_{cristae}, β_{src}, $k_{\mathrm{src,dec}}$, $K_{\mathrm{SRC,mem}}$	0.044, 0.074, 1.32, 0.048, 0.68	h^{-1}, –	[M/F]	Cristae remodeling & $\mathrm{T_{SCM}}$ saturation [2]
Exhaustion E_{TOX}	k_{exh}, θ_{exh}, $k_{\mathrm{warb,exh}}$, $k_{\mathrm{acid,exh}}$, $r_{\mathrm{epig},0}$, $k_{\mathrm{rev,epig}}$	0.0038, 1.36, 0.0032, 0.22, 0.0032, 0.011	h^{-1}, –	[F]	Glycolytic/acidotic TOX^+ commitment vs. remodeling [5]
Perfusate pH / Flush	$k_{\mathrm{perf},0}$, $k_{\mathrm{perf,fl}}$, $k_{\mathrm{perf,LP}}$, $k_{\mathrm{fl,LP}}$, $\mathrm{pH}_{\mathrm{media}}$, k_{acid}, R_{buf}, N_{ref}	0.48, 1.80, 0.12, 0.85, 7.36, 0.007, 12.0, 48.0	h^{-1}, pH, fold	[A]	Hollow-fiber buffer exchange & bypass neutral flush

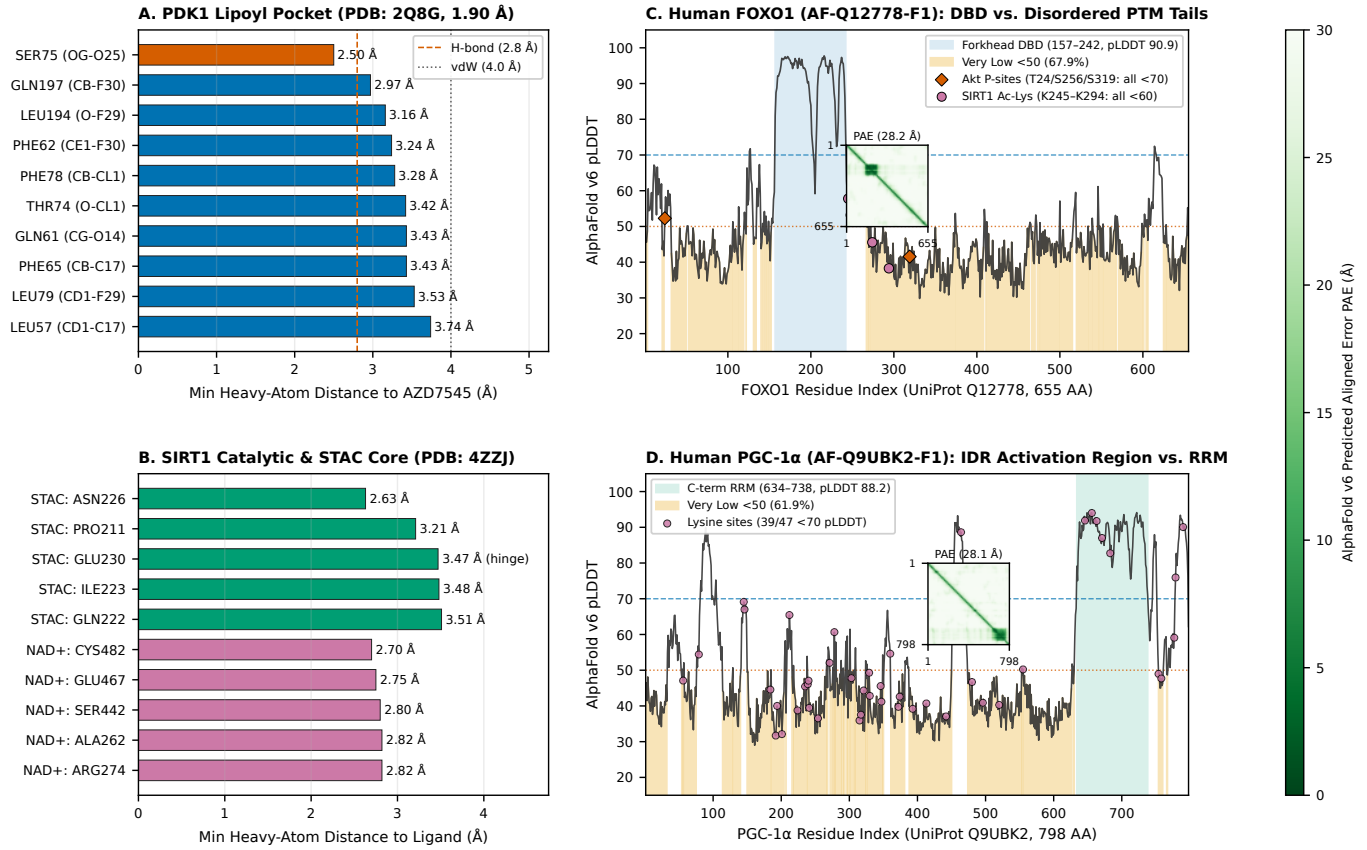


Figure 1: **Atomic Ligand-Contact and AlphaFold v6 pLDDT/PAE Structural Bioinformatics Analysis Guiding Stage-2 Actuator Selection. (A)** Heavy-atom contacts (≤ 4.0 Å) between human `PDK1` and the allosteric inhibitor `AZD7545` (`AZX`) in `PDB: 2Q8G` (1.90 Å X-ray [10]), anchored by a 2.50 Å hydrogen bond at `Ser75` (Oγ–O25) inside the N-terminal lipoyl-binding (`L2`) pocket. **(B)** Heavy-atom contacts (≤ 4.0 Å) in human `SIRT1` (`PDB: 4ZZJ`, 2.74 Å X-ray [15]) to the Sirtuin-Activating Compound `4TQ` (`Asn226` 2.63 Å; allosteric hinge `Glu230` 3.47 Å) and active-site carba-NAD^+ (`CNA`). **(C)** Full-length human `FOXO1` (`AF-Q12778-F1-v6`, 655 AA): only the Forkhead DNA-binding domain (`residues 157-243`, mean pLDDT = 90.62) is structured, whereas 9/9 `Akt` phosphorylation (`Thr24, Ser256, Ser319`) and `SIRT1` deacetylation (`Lys245-Lys294`) sites reside in disordered segments (pLDDT < 70; inset: 655×655 `PAE` matrix, off-diagonal mean 28.23 Å). **(D)** Full-length human `PGC-1`α (`AF-Q9UBK2-F1-v6`, 798 AA): only the C-terminal RNA-recognition motif (`residues 634-738`, mean pLDDT = 88.22) is structured, whereas 39/47 (82.98%) of lysine residues and the entire N-terminal/central activation region (`1-633`) are disordered (pLDDT < 70).

ligand hydroxyl (O25), flanked by `Gln197` (2.97 Å), `Leu194` (3.16 Å), `Phe62` (3.24 Å), `Phe78` (3.28 Å), `Thr74` (3.42 Å), `Gln61` (3.43 Å), `Phe65` (3.43 Å), `Leu79` (3.53 Å), `Leu57` (3.74 Å), `Asp64` (3.92 Å), and `Leu201` (3.92 Å). By severing `PDK1` tethering to the dihydrolipoamide acetyltransferase (`E2`) core of the `PDH` complex, `AZD7545` unblocks pyruvate oxidation without competing against millimolar intramitochondrial ATP.

Similarly, coordinate audit of the engineered human SIRT1 construct bound to Sirtuin-Activating Compound 4TQ and active-site carba-NAD^+ (PDB: 4ZZJ, Chain A, 344 resolved residues within 183-660 at 2.74 Å resolution [15]) resolves two distinct structured binding sites (Figure 1B): (i) the N-terminal three-helix STAC-binding domain, where 4TQ (36 heavy atoms) contacts Asn226 (2.63 Å Nδ2–OAN hydrogen bond), Pro211 (3.21 Å), the critical electrostatic hinge residue Glu230 (3.47 Å Oε1–NAB), Ile223 (3.48 Å), and Gln222 (3.51 Å); and (ii) the Rossmann-fold NAD^+-binding pocket, where carba-NAD^+ (CNA, 44 heavy atoms) is coordinated by Cys482 (2.70 Å), Glu467 (2.75 Å), Ser442 (2.80 Å), Ala262 (2.82 Å), and Arg274 (2.82 Å).

Chemical-Probe Caveat for SIRT1 STACs: Early synthetic STACs such as SRT1720 were shown by Pacholec et al. [12] to exhibit fluorophore-dependent artifacts on TAMRA-labeled peptides and off-target pharmacology in biochemical panels, while Hubbard et al. [13] demonstrated that Glu230-mediated allosteric activation of native substrates requires bulky hydrophobic residues at positions +1/ + 6 C-terminal to the target acetyl-lysine. Consequently, in our model and postulated benchtop falsification protocol (§7), Stage-2 SIRT1 activation (u_{sirt}) primarily represents NAD^+ substrate repletion via nicotinamide riboside (NR, acting directly at the Cys482/Glu467/Arg274 Rossmann pocket), with synthetic STACs treated strictly as an optional tool-compound arm requiring orthogonal genetic (SIRT1 CRISPR knockout) and Glu230Lys point-mutant controls.

3.2 Domain Architecture and PTM-Tail Disorder of FOXO1 and PGC-1α

Full-length AlphaFold v6 structural audit reveals a sharp dichotomy between isolated nucleic-acid-binding domains and disordered post-translational modification (PTM) regions in both downstream effectors (Figure 1C–D, Table 2):

1. **Human FOXO1 (AF-Q12778-F1-v6, 655 AA):** Global mean pLDDT is 50.96, with 67.94% (445/655 AA) in the *Very Low* (< 50) band, 18.17% (119/655) in 50–70, and 13.89% (91/655) $\geq$ 70, accompanied by a high off-diagonal mean PAE = 28.23 Å (1.86% residue pairs < 5 Å). Contiguous domain segmentation ($\geq$ 25 AA with pLDDT $\geq$ 70) identifies *one* structured domain: the winged-helix Forkhead DNA-binding domain (residues 157-243, 87 AA, mean pLDDT = 90.62, internal PAE = 4.70 Å). Crucially, however, **all 9/9 canonical regulatory PTM residues** controlling nuclear-cytoplasmic shuttling lie outside rigid tertiary pockets (pLDDT < 70): the three Akt 14-3-3 phosphosites (Thr24: 52.28, Ser256: 66.75, Ser319: 41.56) and all six CBP/p300/SIRT1 acetyl-lysines (Lys245: 57.78, Lys248: 53.19, Lys262: 55.19, Lys265: 52.41, Lys274: 45.56, Lys294: 38.28).
2. **Human PGC-1α (AF-Q9UBK2-F1-v6, 798 AA):** Global mean pLDDT is 52.74 (61.90% < 50, 17.17% in 50–70, 20.93% $\geq$ 70; off-diagonal mean PAE = 28.12 Å, 1.26% pairs < 5 Å). Segmentation detects *one* structured C-terminal RNA-recognition motif (RRM, residues 634-738, 105 AA, mean pLDDT = 88.22, internal PAE = 7.79 Å), whereas the entire 633-residue N-terminal and central transcriptional coactivation region—and 39/47 (82.98%) of all lysine residues across the protein (mean lysine pLDDT = 53.53)—is intrinsically disordered.

Rather than claiming that disordered transcription factors are universally "undruggable," this structural audit explains *mechanistically* why conventional pocket-based small-molecule discovery lacks preformed allosteric cavities in the PTM control regions of FOXO1 and PGC-1α, whereas the structured catalytic cleft of SIRT1 (PDB: 4ZZJ) is evolutionarily specialized to capture flexible acetyl-lysine peptide motifs.

4 In Silico Protocol Sweep, Target Ablations, and Adjoint Pareto Optimality

4.1 Nominal Switching Sweep and Mechanistic Component Ablations

To evaluate whether two-stage metabolic gating resolves the simulated proliferation–exhaustion conflict, we integrated the nominal 8-state system (RK4, N = 1,000 steps, Δt = 0.168 h) across 14 open-loop, closed-loop, and ablated regimens against five simultaneous manufacturing release criteria: viable expansion $N_{\text{cells}}(168\text{ h}) \geq 50.0\times$, stemness $\text{T}_{\text{SCM}}(168\text{ h}) \geq 68.0\%$, exhaustion $E_{\text{TOX}}(168\text{ h}) \leq 15.0\%$, SRC $\geq 3.5\times$, and depolarized organelle fraction $h(168\text{ h}) \leq 10.0\%$ (Figure 2A–B, Table 3).

Unphased Warburg expansion ($u_1 = u_2 = 0$) achieves maximal cell yield (81.77×) but drives severe mitochondrial depolarization (h = 66.76%), supernatant lactatemia (R_{LP} = 30.02, pH = 6.92), nuclear FOXO1 exclusion (F_{nuc} = 30.48%), TOX^+ commitment (76.13%), and T_{SCM} collapse (3.67%). Conversely, premature oxidative switching at Hour 0 ($u_1 = 0, u_2 = 1$) stalls clonal division at 9.58× (< 50× floor) while failing to clear baseline depolarized mitochondria (h = 64.81%). Continuous urolithin A monotherapy ($u_1 = 1, u_2 = 0$, without Stage-2 galactose/AZD7545/SIRT1 consolidation) thoroughly purges depolarized mitochondria (h = 0.04%) and preserves blastogenesis (80.60×), but because high glycolytic flux continues to drive Warburg exhaustion (41.03%) and prevents full cristae SRC maturation (2.26×), simulated T_{SCM} plateaus at 31.10%.

Table 2: **Verified Atomic-Coordinate and AlphaFold v6 Structural Bioinformatics Summary.** All metrics are computed directly from local `mmCIF` and `PAE` JSON coordinate files by `audit_all_structures()` in `simulator/generate_paper2_verified_benchmarks.py`.

Target Protein	Structure / Accession	Length / Res.	Structured Domain(s)	Disorder / PAE Metrics	Verified Key Contacts or PTM Sites
PDK1 (Kinase)	PDB: 2Q8G (X-ray)	368 AA (1.90 Å)	Lipoyl (L2) + Kinase core	Rigid co-crystal (41–423)	AZD7545: Ser75 (2.50 Å), Gln197 (2.97 Å), Leu194 (3.16 Å)
SIRT1 (Deacetylase)	PDB: 4ZZJ (X-ray)	344 AA (2.74 Å)	STAC 3-helix + Rossmann	Rigid co-crystal (183–660)	4TQ: Asn226 (2.63 Å), Glu230 (3.47 Å); CNA: Cys482 (2.70 Å)
FOXO1 (TF)	AF-Q12778-F1-v6	655 AA (AF v6)	Forkhead DBD 157–243 (90.6)	67.9% < 50; PAE = 28.23 Å	9/9 Akt (T24/S256/S319) & Ac-Lys (K245–K294) < 70 pLDDT
PGC-1α (Coact.)	AF-Q9UBK2-F1-v6	798 AA (AF v6)	C-term RRM 634–738 (88.2)	61.9% < 50; PAE = 28.12 Å	1–633 activation IDR; 39/47 (83.0%) lysines < 70 pLDDT

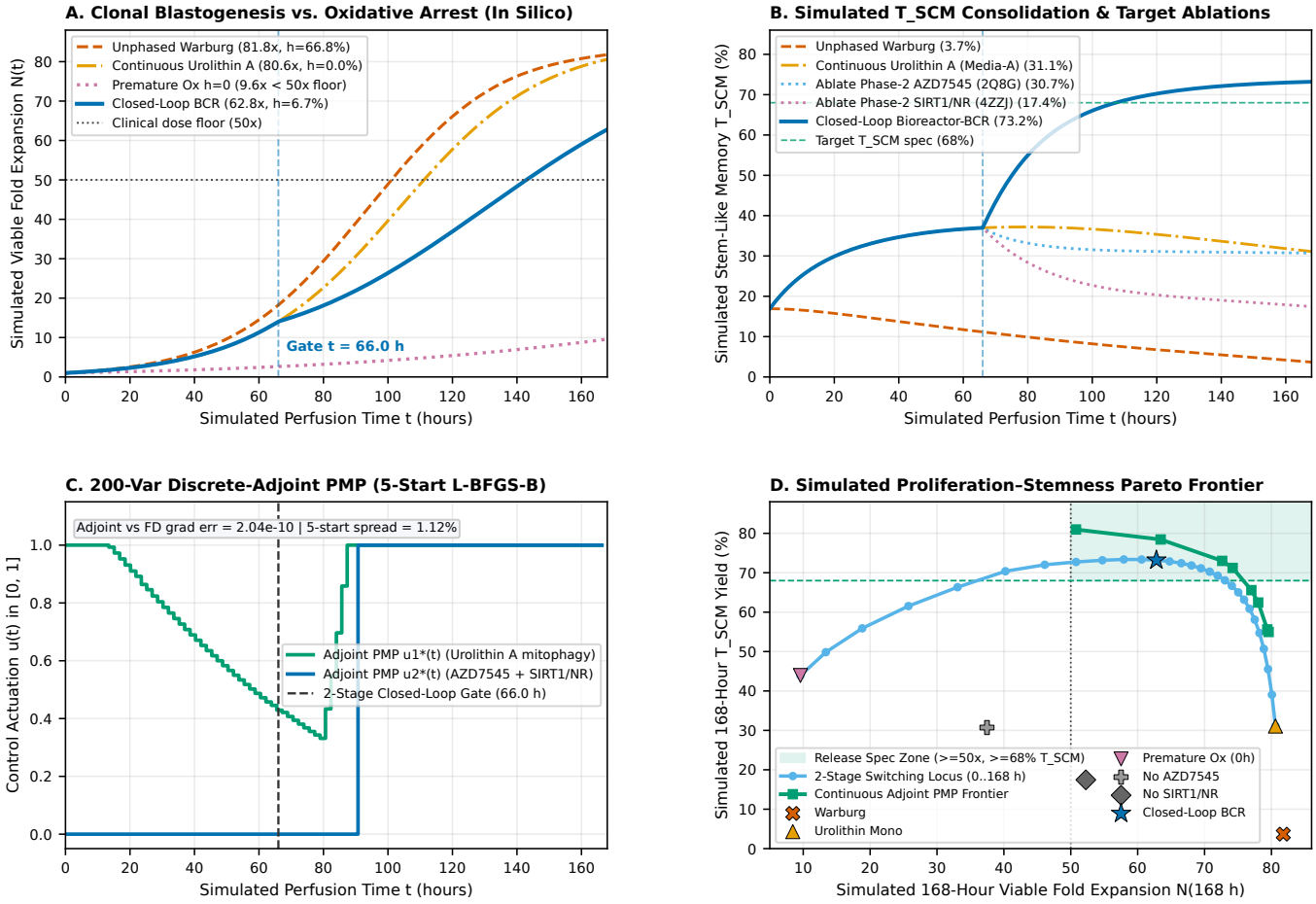


Figure 2: ***In Silico* 168-Hour Perfusion Trajectories, Component Ablations, and 200-Variable Discrete-Adjoint Pareto Frontier.** **(A)** Simulated clonal fold expansion $N_{\text{cells}}(t)$ showing Warburg glycolytic expansion (81.77×, h = 66.76%), premature Day-0 oxidative forcing (9.58× < 50× clinical floor), continuous urolithin A monotherapy (80.60×), and closed-loop **Bioreactor-BCR** switching at t_{gate} = 66.02 h (62.80×, h = 6.73%). **(B)** Simulated $\text{T}_{\text{SCM}}(t)$ consolidation comparing closed-loop gating (73.20%) against continuous urolithin A without Stage-2 oxidative consolidation (31.10%) and single-target Stage-2 ablations of `AZD7545` (30.72%) or `SIRT1/NR` (17.45%). **(C)** Unconstrained 200-variable discrete-adjoint Pontryagin optimal control schedule $(u_1^*(t), u_2^*(t)$ across 100 intervals; maximum adjoint vs. central finite-difference gradient error across all 200 variables 2.04×10^{-10}, 5-start `L-BFGS-B` objective spread 1.12%). **(D)** Simulated proliferation–stemness Pareto frontier comparing continuous PMP modulation against the 2-stage switching locus ($t_{\text{sw}} \in [0, 168$ h]) and discrete baseline/ablation protocols.

Under nominal closed-loop gating ($h(t) \leq 0.045$, $N_{\text{cells}}(t) \geq 14.0\times$, $\text{pH}(t) \geq 7.10$), **Bioreactor-BCR** transitions from Media-A to Media-B at t_{gate} = 66.02 h (accumulated selection integral $s(t_{\text{gate}}) = 2.8248$), achieving 62.80× fold expansion, 73.20% T_{SCM}, SRC = 4.87×, E_{TOX} = 11.75%, and h(168 h) = 6.73%—satisfying all five release criteria simultaneously. Ablating Stage-2 `AZD7545` ($u_{\text{azd}} = 0$ after 66.02 h) leaves `PDH` phosphorylated ($g_{\text{pdh}} = 0.15$), cutting expansion in galactose-shifted media to 37.47× and T_{SCM} to 30.72%. Ablating Stage-2 `SIRT1/NR` activation ($u_{\text{sirt}} = 0$ after 66.02 h) allows D to out-replicate W ($k_{\text{rep,adv}} = 1.22$), causing h to rebound

from 4.50% at the gate to 36.27% at 168 h and collapsing T_{SCM} to 17.45%.

4.2 200-Variable Discrete-Adjoint Optimal Control and Pareto Frontier

To test whether a simple two-stage pinch-valve switch sacrifices substantial performance relative to an unconstrained, continuously modulated multi-pump schedule $(u_1(t), u_2(t)) \in [0,1]^2$, we discretized $[0, 168\text{ h}]$ into $N_{ocp} = 100$ piecewise-constant control intervals (200 decision variables) and minimized the exact scalarized terminal-plus-effort Bolza objective implemented in `solve_discrete_adjoint_pareto()`:

$$\begin{aligned} J_\alpha(\mathbf{u}) = &-\alpha\frac{T_{SCM}(\mathbf{x}(T))}{100} - (1-\alpha)\frac{N_{cells}(\mathbf{x}(T))}{75.0} \\ &+ 0.45\Big(\frac{h(T)}{0.10}\Big)^2 + 0.35\,E_{TOX}(T)^2 \\ &+ 2.50\Big(\frac{\max_\epsilon(0,\ 50.0 - N_{cells}(T))}{50.0}\Big)^2 \\ &+ \frac{0.008}{N_{ocp}}\sum_{k=0}^{N_{ocp}-1}\big(u_{1,k}^2 + u_{2,k}^2\big), \end{aligned} \tag{22}$$

where $\max_\epsilon(0,z) \equiv \frac{1}{2}(z + \sqrt{z^2+\epsilon^2})$ ($\epsilon = 0.25$ on the 50× dose floor; smooth hinge $\epsilon = 10^{-3}$ inside the differentiable discrete-adjoint ODE RHS, evaluated on the exact non-smooth `RK4` plant at $N = 1{,}000$ fine steps for Table 3). Propagating exact discrete-time `RK4` stage-by-stage Jacobians backward via $\mathbf{p}_k = (\partial\Phi_k/\partial\mathbf{x}_k)^\top \mathbf{p}_{k+1}$ ($\mathbf{p}_k \in \mathbb{R}^8$) and verifying against central finite differences ($\epsilon_{FD} = 10^{-6}$) across *all* 200 *decision variables* yields a maximum gradient discrepancy of 2.04×10^{-10}. Launching `L-BFGS-B` at the balanced Pareto weight $\alpha = 0.58$ from 5 diverse initializations (two-stage step at 66.0 h, uniform interior $u_1 = u_2 = 0.5$, early step at 36.0 h, late step at 96.0 h, and pseudo-random interior schedule $\mathcal{U}[0.1, 0.9]$) converges to a tight objective spread of 1.12% ($J^*_{0.58} = -0.81366$ vs. two-stage closed-loop $J_{0.58} = -0.58290$, Table 3).

Interestingly, the Pontryagin-optimal control profile (Figure 2C) independently discovers a *two-phase temporal structure*: it holds $u_2^*(t) = 0$ throughout Hours 0–90.7 while modulating $u_1^*(t)$ along a smooth interior arc ($1.0 \to 0.33$, avoiding the small c_{akt} glycolytic proliferation penalty once $h(t)$ is low), executes a brief 15-hour co-actuation overlap ($u_1^* = 1, u_2^* = 0 \to 1$) to block $k_{rep,adv}$ rebound during initial mitochondrial biogenesis, and holds $u_1^* = u_2^* = 1$ to Hour 168. Sweeping $\alpha \in [0.08, 0.92]$ (Figure 2D) shows that while continuous multi-pump modulation expands yield from 62.80× to 71.15× (+1.26% T_{SCM}), the single-valve two-stage locus already enters well inside the 5-criteria clinical release zone without requiring continuous microfluidic drug mixing.

5 Heterogeneous Oncology Cohort, Ratiometric `ORR` Observability, and Adaptive Extended-Horizon Rescue

5.1 Volumetric Respirometry Bias vs. Cell-Density Cancellation in Ratiometric `ORR`

In a physical closed-cassette perfusion bioreactor, inline dissolved-O_2 optodes do *not* measure per-cell specific respiration directly; they measure *volumetric* oxygen depletion proportional to total viable cell count N_{cells} times per-cell oxidative flux, while inline near-infrared (`NIR`) backscatter or capacitance probes estimate cell concentration subject to donor-specific lymphoblast-size calibration bias α_{OD}. To evaluate observability under realistic bioprocess physics and statistical power, we scaled our virtual cohort to $N = 1{,}000$ oncology apheresis donors ($h_0 \sim \mathcal{U}[0.20, 0.62]$, $M_0 \sim \mathcal{U}[0.85, 1.12]$, $R_{LP,0} \sim \mathcal{U}[18, 28]$, $F_{nuc,0} \sim \mathcal{U}[0.22, 0.36]$, $S_{SRC,0} \sim \mathcal{U}[1.05, 1.50]$, $E_{TOX,0} \sim \mathcal{U}[0.05, 0.14]$) with unmodeled kinetic perturbations ($\mu_{glyc} \times \mathcal{U}[0.66, 1.36]$, $\mu_{ox} \times \mathcal{U}[0.84, 1.16]$, $k_{mito,D,max} \times \mathcal{U}[0.68, 1.36]$, $k_{Warburg} \times \mathcal{U}[0.82, 1.22]$, $k_{warb,exh} \times \mathcal{U}[0.82, 1.22]$) and unmodeled ±6% optical-density scattering bias $\alpha_{OD} \sim \mathcal{U}[0.94, 1.06]$. The Huber-EKF runs strictly on `NOMINAL_PARAMS` ($\hat{\alpha}_{OD} = 1.00$) and never observes any donor's private parameters or initial state.

Sampled every $\Delta t_{obs} = 0.30$ h (18 min) under correlated AR(1) noise ($\rho = 0.65$), we compare:

1. **Physical 4-Channel Cassette Suite (No `ORR`):** Amperometric supernatant ratio $y_1 = R_{LP} + \nu_1$ ($\sigma_1 = 0.55$), *volumetric* optode oxygen uptake $y_2 = OUR_{vol}(\mathbf{x}, u_2) + \nu_2 = N_{cells}\,ATP_{ox}(\mathbf{x}, u_2) + \nu_2$ (heteroscedastic $\sigma_{2,k} = 0.030\sqrt{1 + (0.08N_{cells})^2}$), uncalibrated NIR optical density $y_3 = \alpha_{OD}N_{cells} + \nu_3$ ($\sigma_{3,k} = \sqrt{0.35^2 + (0.025N_{cells})^2}$), and phenol-red optical transmission $y_4 = \text{pH} + \nu_4$ ($\sigma_4 = 0.018$).

2. **5-Channel Optical-Metabolic Suite (With Ratiometric `ORR`):** Appends quartz-window two-band FAD/(FAD + NAD(P)H) autofluorescence [14]. Because bulk suspension fluorescence in both spectral bands scales linearly with illuminated cell concentration ($I_{FAD} = \beta_{opt}N_{cells}W$, $I_{NAD(P)H} = \beta_{opt}N_{cells}[D + c_{glyc}g_{glc}(u_2)]$), the unknown factor $\beta_{opt}N_{cells}$ *cancels identically* in the ratio:

$$\begin{aligned} y_5 &= \frac{I_{FAD}}{I_{FAD} + I_{NAD(P)H}} + \nu_5 \\ &= \frac{W}{W + D + c_{glyc}\,g_{glc}(u_2)} + \nu_5, \end{aligned} \tag{23}$$

with $\sigma_5 = 0.012$, nominal $c_{glyc} = 0.35$, and ±8% unmodeled donor variation in c_{glyc}.

Table 3: **Nominal 168-Hour *In Silico* Protocol Sweep, Target Ablations, and Discrete-Adjoint PMP Benchmark.** Release specification: $N \geq 50\times$, $\mathrm{T_{SCM}} \geq 68\%$, $\mathrm{TOX^+} \leq 15\%$, $\mathrm{SRC} \geq 3.5\times$, and $h \leq 10\%$.

Simulated Regimen	t_{gate} (h)	$s(t_{\mathrm{gate}})$	**Fold** N(168 h)	$\mathrm{T_{SCM}}$ (%)	$\mathrm{TOX^+}$ (%)	SRC (×)	h(168 h) (%)	**5-Spec Pass**
Unphased Warburg ($u_1 = 0, u_2 = 0$)	–	–	81.77×	3.67%	76.13%	0.69×	66.76%	Fail (4/5)
Premature Oxidative ($u_1 = 0, u_2 = 1$ from 0 h)	0.00	0.0000	9.58×	43.92%	27.64%	1.22×	64.81%	Fail (5/5)
Continuous Urolithin A Monotherapy ($u_1 = 1, u_2 = 0$)	–	–	80.60×	31.10%	41.03%	2.26×	0.04%	Fail (3/5)
Fixed Early Switch ($t_{\mathrm{sw}} = 36$ h)	36.12	1.4696	46.11×	72.02%	9.32%	3.58×	23.51%	Fail (N, h)
Fixed Intermediate Switch ($t_{\mathrm{sw}} = 48$ h)	48.05	2.0009	54.72×	73.17%	10.16%	4.23×	14.87%	Fail ($h > 10\%$)
Closed-Loop Bioreactor-BCR ($h \leq 4.5\%$)	**66.02**	**2.8248**	**62.80×**	**73.20%**	**11.75%**	**4.87×**	**6.73%**	**PASS** (5/5)
Fixed Late Switch ($t_{\mathrm{sw}} = 96$ h)	96.10	4.2392	70.77×	70.26%	15.77%	5.13×	1.54%	Fail ($\mathrm{TOX^+}$)
Fixed Very Late Switch ($t_{\mathrm{sw}} = 120$ h)	120.12	5.3857	75.04×	65.03%	21.08%	4.78×	0.46%	Fail ($\mathrm{T_{SCM}}, \mathrm{TOX^+}$)
Ablate Stage-2 `AZD7545` (`PDB: 2Q8G`, $u_{\mathrm{azd}} = 0$)	66.02	2.8248	37.47×	30.72%	21.25%	2.66×	10.82%	Fail (5/5)
Ablate Stage-2 `SIRT1/NR` (`PDB: 4ZZJ`, $u_{\mathrm{sirt}} = 0$)	66.02	2.8248	52.25×	17.45%	28.46%	1.62×	36.27%	Fail (4/5)
Discrete-Adjoint PMP Optimal (200 vars, $\alpha = 0.58$)	Cont.	–	**71.15×**	**74.46%**	**14.08%**	**5.33×**	**0.09%**	**PASS** (5/5)

In contrast, dividing volumetric respiration y_2 by biased optical density y_3 yields $y_2/y_3 \approx \alpha_{\mathrm{OD}}^{-1}(W + \eta_D D)(1 + \gamma_{\mathrm{pdh}} u_2)(1 + \gamma_{\mathrm{src}}(S_{\mathrm{SRC}} - 1)_+)$, which couples only to the sum $W + \eta_D D$ and inherits the permanent $\alpha_{\mathrm{OD}}^{-1}$ lymphoblast-size bias. Across the six core gating states $[W, D, N_{\mathrm{cells}}, R_{\mathrm{LP}}, S_{\mathrm{SRC}}, \mathrm{pH}]$, the 64-hour empirical Observability Gramian reflects this physical bottleneck: without `ORR`, $\kappa(\mathcal{G}_{4\mathrm{ch}}) = 5.37 \times 10^5$ and degrades 13.4× upon R_{LP} electrode fouling to $\kappa(\mathcal{G}_{3\mathrm{ch,foul}}) = 7.21 \times 10^6$ ($\sigma_{\min} = 1.34 \times 10^{-5}$), yielding $\hat{h}$ RMSE = 6.48% (unfouled) and 16.18% (biofouled). Adding ratiometric `ORR` (23) supplies an orthogonal sensitivity row independent of N_{cells} and α_{OD}, improving post-fouling Gramian conditioning 8.55× ($\kappa = 8.43 \times 10^5$) and holding $\hat{h}$ RMSE to 2.08% (unfouled) and 2.15% (biofouled).

5.2 Explicit Switching Law, Guard Ablation, and $N = 1{,}000$ Paired McNemar Benchmarks

To ensure complete controller transparency between Python and C99, the closed-loop Stage-1 → 2 transition requires 4 consecutive frames (1.2 h persistence quorum) with $y_4 \geq 7.10$ and satisfaction of either the primary reachability gate or an explicit late-glycolysis exhaustion guard:

$$\mathcal{C}_{\mathrm{switch}}(t) = \big(\hat{h} \leq 0.056 \wedge \hat{N}_{\mathrm{cells}} \geq N_{\mathrm{req}}(t, \hat{\mu}_{\mathrm{rel}})\big) \vee \underbrace{\big(t \geq 83 \text{ h} \wedge \hat{h} \leq 0.072 \wedge \hat{N}_{\mathrm{cells}} \geq N_{\mathrm{guard}}\big)}_{\text{Late-glycolysis } \mathrm{TOX^+} \text{ ceiling guard}}, \tag{24}$$

where $\hat{\mu}_{\mathrm{rel}}(t) = \mathrm{clip}\big(\ln(\hat{N}_{\mathrm{cells}}(t)/\hat{N}_{\mathrm{cells}}(24\text{ h}))/[0.044(t - 24)], 0.78, 1.22\big)$ for $t \geq 28$ h (1.0 for $t < 28$ h), predicted Stage-2 expansion factor $M_{\mathrm{S2}}(t) = \exp\big(0.0122\sqrt{\hat{\mu}_{\mathrm{rel}}(t)}\,(T_{\mathrm{plan}} - t)_+\big)$, and $N_{\mathrm{req}}(t, \hat{\mu}_{\mathrm{rel}}) = \mathrm{clip}\big(N_{\mathrm{target}}/\max(1.05, 0.58 M_{\mathrm{S2}} + 0.42\min(M_{\mathrm{S2}}, 3.6)), 14.0, 42.0\big)$ ($T_{\mathrm{plan}} = 168$ h, $N_{\mathrm{target}} = 52.5\times$ to buffer ±6% `OD` bias above the true 50× floor, $N_{\mathrm{guard}} = 18.0\times$).

Across all $N = 1{,}000$ oncology donors (Figure 3A, Table 4), unphased Warburg, premature Hour-0 oxidative switching, and continuous urolithin A monotherapy achieve 0.0% (0/1,000) 5-criteria pass rates. Among nine open-loop fixed timers (24–120 h), performance peaks sharply at $t_{\mathrm{sw}} = 68$ h (54.1%, 541/1,000) and 76 h (52.8%). Because Warburg-flux and `OD` biases corrupt $\hat{h}$ without `ORR`, the 4-channel Huber-EKF achieves only 40.5% (405/1,000). The 5-channel `ORR` Huber-EKF raises the 168-hour pass rate to 58.7% (587/1,000), significantly outperforming both the retrospective-best 68 h fixed timer (+4.6%, paired 95% CI [+1.97%, +7.23%], exact two-sided McNemar $p = 8.04 \times 10^{-4}$; 114 discordant wins vs. 68 losses) and the 4-channel No-`ORR` EKF (+18.2%, 95% CI [+15.58%, +20.82%], $p = 1.19 \times 10^{-41}$; 197 vs. 15). Ablating the $t \geq 83.0$ h exhaustion guard in (24) drops the 5-channel pass rate from 58.7% to 49.9% (−8.8%, 95% CI [−10.58%, −7.02%], $p = 1.47 \times 10^{-25}$), confirming that capping glycolytic dwell time is essential to prevent $\mathrm{TOX^+} > 15\%$ in high-h_0 donors. Under 100% amperometric `Lac:Pyr` biofouling ($t_{\mathrm{foul}} \sim \mathcal{U}[30, 48]$ h), naive `Lac:Pyr` gating collapses to 0.5% (5/1,000) and 4-channel NIS handover yields 12.3% (123/1,000), whereas 5-channel `ORR`+NIS handover ($\mathrm{NIS}_1 > 13.5$ for 3 frames, 1,000/1,000 handovers) preserves 59.4% (594/1,000) (+47.1% vs. No-`ORR` handover, $p = 5.22 \times 10^{-129}$; +5.3% vs. unfouled 68 h fixed timer, $p = 1.81 \times 10^{-4}$).

5.3 Direct Test of Extended-Horizon (168–240 h) Rescue for Hypoproliferative Donors

Evaluating the full-information donor oracle at 168 h shows that 74.3% (743/1,000) of the cohort is controllable within 7 days, leaving 25.7% (257/1,000) infeasible at fixed 168 h (median $h_0 = 51.19\%$, $\mu_{\mathrm{glyc}}/\mu_{\mathrm{nom}} = 0.773\times$ vs. 38.79% and 1.059× in controllable donors). Rather than leaving extended culture as an untested conjecture, we simulated an adaptive variable-harvest policy ($T_{\mathrm{plan}} = 240$ h, $N_{\mathrm{target}} = 53.5\times$, $N_{\mathrm{guard}} = 14.0\times$) that harvests at 168 h whenever $\hat{N}_{\mathrm{cells}}(168\text{ h}) \geq 53.5\times$ with ≥ 36 h Stage-2 consolidation, and otherwise continues Stage-2 perfusion up to 240 h until $\hat{N}_{\mathrm{cells}}(t) \geq 53.5\times$. Extending the horizon raises the full-information oracle ceiling from 74.3% to 97.3% (973/1,000)—proving that 89.5% (230/257) of 168 h-infeasible runs fail solely from division-time truncation, while only 2.7% (27/1,000; median $h_0 = 57.14\%$,

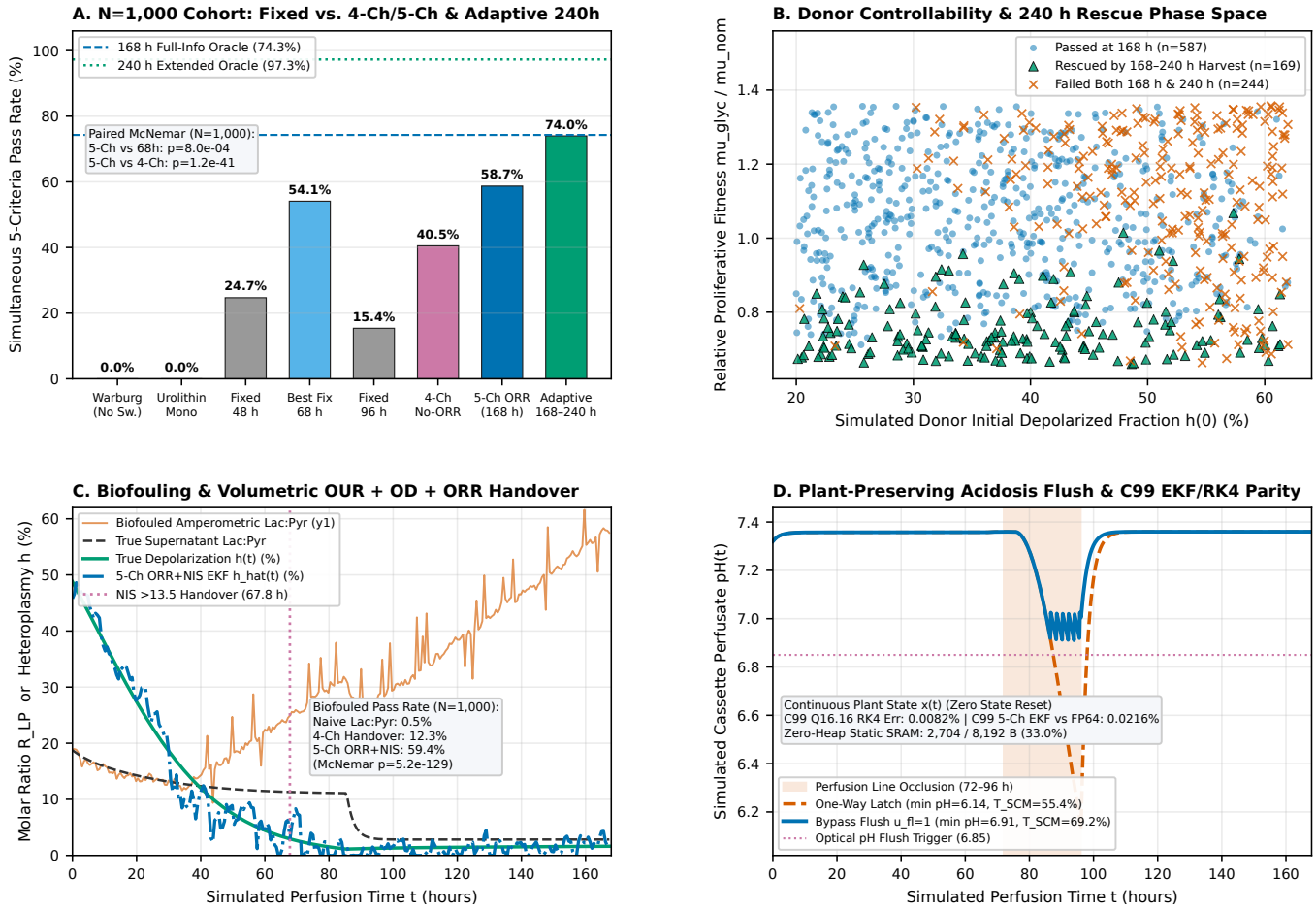


Figure 3: $N = 1{,}000$ **Heterogeneous Oncology Donor Cohort, Physical Volumetric `OUR`/`OD`/`ORR` Observability Under Biofouling, and Extended-Horizon Rescue. (A)** Simultaneous 5-criteria manufacturing pass rates across $N = 1{,}000$ simulated oncology apheresis donors comparing unphased baselines (0.0%), open-loop fixed timers (peaking at 54.1% at 68 h), physical 4-channel EKF without `ORR` (40.5%), 5-channel `ORR`-augmented Huber-EKF at 168 h (58.7%; exact paired McNemar $p = 8.0 \times 10^{-4}$ vs. 68 h, $p = 1.2 \times 10^{-41}$ vs. 4-channel), and `OD`-triggered adaptive harvest (168–240 h, 74.0%; 168 h oracle 74.3%, 240 h oracle 97.3%). **(B)** 2D oncology donor controllability phase space $(h_0, \mu_{\text{glyc}}/\mu_{\text{nom}})$ partitioning the 1,000 donors into 168 h passers ($n = 587$), hypoproliferative donors rescued by 168–240 h adaptive harvest ($n = 169$), and unrescued severe-burden donors ($n = 244$). **(C)** Representative biofouling trajectory where enzymatic `Lac:Pyr` drift triggers NIS > 13.5 residual handover at 67.8 h to the optical quartet (`OUR`$_{\text{vol}}$+`OD`+`pH`+`ORR`), preserving a 59.4% cohort pass rate ($p = 5.2 \times 10^{-129}$ vs. 12.3% 4-channel handover). **(D)** Plant-preserving bypass neutral flush ($u_{\text{fl}} = 1$, zero state reset) during perfusion-line occlusion (72–96 h), lifting min pH from 6.14 to 6.91, alongside freestanding C99 `Q16.16 RK4` (0.0082%) and 5-channel EKF (0.0216%) parity.

fast Warburg exhaustion $\mu_{\text{glyc}}/\mu_{\text{nom}} = 1.189\times$) are intrinsically refractory even at 240 h. In closed loop, `OD`-guided adaptive harvest extends culture past 168 h in only 22.5% of donors yet rescues 169 previously failing hypoproliferative runs (green triangles in Figure 3B), lifting the closed-loop pass rate from 58.7% to 74.0% (740/1,000) (+15.3%, 95% CI [+12.81%, +17.79%], exact McNemar $p = 2.08 \times 10^{-33}$).

6 Plant-Preserving Acidosis Rescue and Deterministic Fixed-Point Safety Firmware

6.1 Continuous-Plant Neutral Bypass Flush Under Perfusion Occlusion

As established in Paper I [11], restoring a microcontroller's internal estimator memory after a fault cannot reset biological state variables; actuator recovery must operate strictly through continuous plant dynamics $\mathbf{x}(t)$. In hollow-fiber CAR-T cassettes, a primary post-switch haz-

Table 4: **Simulated $N = 1{,}000$ Heterogeneous Oncology Apheresis Cohort, Guard/Sensor Ablations, Biofouling Handover, and Extended-Horizon (168–240 h) Rescue.** Medians and interquartile ranges (IQR) across all 1,000 donors (seed `20261111`).

Cohort Control Policy ($N = 1{,}000$ Donors)	Pass Rate	Fold N Med [IQR]	T_{SCM} Med [IQR]	TOX^+ Med	h_{end} Med	$\hat{h}$ RMSE	Gate t_{sw} Med [IQR]
Unphased Warburg ($u_1 = 0, u_2 = 0$, 168 h)	0.0% (0/1000)	80.7× [74.6, 83.7]	4.1% [1.8, 6.7]	75.0%	63.7%	–	–
Continuous Urolithin A Monotherapy (168 h)	0.0% (0/1000)	81.3× [72.2, 83.9]	30.5% [28.9, 32.1]	42.0%	0.03%	–	–
Open-Loop Fixed Timer (48 h)	24.7% (247/1000)	55.7× [42.1, 66.1]	72.9% [71.1, 74.2]	10.9%	13.1%	–	48.0 h (fixed)
Open-Loop Fixed Timer (60 h)	45.0% (450/1000)	61.8× [47.2, 71.6]	72.9% [71.5, 73.9]	11.9%	7.5%	–	60.0 h (fixed)
Best Open-Loop Fixed Timer (68 h)	**54.1%** (541/1000)	64.9× [50.3, 74.3]	72.5% [71.3, 73.6]	12.7%	5.1%	–	68.0 h (fixed)
Open-Loop Fixed Timer (84 h)	42.2% (422/1000)	69.5× [55.0, 77.5]	71.2% [70.0, 72.2]	14.6%	2.3%	–	84.0 h (fixed)
Open-Loop Fixed Timer (96 h)	15.4% (154/1000)	72.1× [58.1, 79.2]	69.5% [68.3, 70.7]	16.5%	1.2%	–	96.0 h (fixed)
Ablation: Physical 4-Ch EKF (**No ORR**, 168 h)	40.5% (405/1000)	68.9× [58.4, 75.8]	70.7% [66.2, 72.4]	14.9%	1.81%	6.48%	84.6 h [66.5, 100.7]
Ablation: 5-Ch ORR EKF (**No 83 h Guard**, 168 h)	49.9% (499/1000)	66.5× [58.4, 73.1]	71.1% [68.3, 72.7]	14.3%	3.81%	2.12%	79.4 h [63.0, 102.3]
Closed-Loop 5-Ch ORR Huber-EKF (168 h)	**58.7%** (587/1000)†	66.4× [56.2, 73.1]	71.7% [70.2, 72.8]	13.7%	4.34%	**2.08%**	79.4 h [63.0, 90.6]
Biofouled Lac:Pyr: Naive Threshold Gate	0.5% (5/1000)	81.2× [72.2, 83.9]	30.5% [29.0, 32.1]	42.0%	0.03%	47.26%	50.7 h ($n = 5$)
Biofouled Lac:Pyr: 4-Ch No-ORR NIS Handover	12.3% (123/1000)	74.2× [62.6, 80.8]	65.2% [60.3, 69.0]	20.9%	0.57%	16.18%	115.8 h [99.9, 129.9]
Biofouled Lac:Pyr: 5-Ch ORR + NIS Handover	**59.4%** (594/1000)‡	67.0× [56.0, 73.5]	71.6% [70.2, 72.8]	13.9%	3.99%	**2.15%**	82.1 h [64.5, 90.6]
Adaptive Horizon (168–240 h) 5-Ch ORR EKF	**74.0%** (740/1000)§	65.7× [55.3, 73.1]	72.4% [71.2, 73.4]	13.1%	4.87%	**2.06%**	73.8 h [62.0, 84.6]
Full-Information Donor Oracle Ceilings (168 h / 240 h)	74.3% (743/1000) at fixed 168 h; 97.3% (973/1000) with 168–240 h extended harvest (2.7% intrinsically refractory)						

Exact two-sided paired McNemar tests: †$p = 8.04 \times 10^{-4}$ vs. 68 h fixed (+4.6%), $p = 1.19 \times 10^{-41}$ vs. 4-Ch No-ORR (+18.2%); ‡$p = 5.22 \times 10^{-129}$ vs. 4-Ch handover; §$p = 2.08 \times 10^{-33}$ vs. 168 h EKF.

ard is partial peristaltic line occlusion or gas-transfer fouling during high-density Stage-2 expansion ($N_{cells} > 20\times$). We simulated a 24-hour Media-B perfusion-line occlusion across Hours 72–96 (92% drop in main-line exchange $k_{perf,0}$ and $k_{perf,LP}$, loss of `AZD7545` delivery $u_{azd,eff} = 0$, and hypoxic glycolytic surge $1.65 \times k_{acid}$, Figure 3D).

Under a one-way valve latch ($u_{fl} = 0$), cassette perfusate pH(t) plunges to min pH = 6.14, triggering severe acidotic `FOXO1` nuclear export (15) and TOX^+ commitment (17) ($E_{TOX} = 32.28\%$, $T_{SCM} = 55.39\%$). When the supervisory controller monitors optical phenol-red pH(t) ≤ 6.85 and opens the auxiliary neutral-buffer bypass flush ($u_{fl} = 1$, reverting to protective Stage-1 media for a 6-hour stabilization window without altering $\mathbf{x}(t)$ discontinuously), min pH is arrested at 6.91, limiting E_{TOX} to 16.11% and rescuing simulated T_{SCM} to 69.16% ($N_{cells} = 70.59\times$, $h = 8.14\%$).

6.2 Freestanding FPU-Free C99 `RK4` and 5-Channel Huber-EKF Verification

Unlike implantable neural devices or solar agricultural sensors, a benchtop GMP cell-therapy bioreactor is wall-powered; embedded microwatt sleep budgets are not a bioprocess bottleneck. Instead, regulatory software verification for single-use disposable cassette controllers (`IEC 62304` safety class) prioritizes *deterministic execution without dynamic heap allocation (`malloc`), absence of non-deterministic floating-point library (`<math.h>`) or FPU exceptions, and statically bounded SRAM residency* portable to FPU-less safety microcontrollers and integer RISC-V cores (`RV32IM/Zve32x`). We therefore implemented the complete 8-state cassette plant, ΣΔ `RK4` predictor, 5-channel sequential Huber-EKF, integer NIS biofouling monitor, and acidosis bypass-flush state machine in freestanding integer C99 (`build/coralnpu/cart_bioreactor_hil_testbench.c`, compiled under `-O2 -Wall -Wextra -Werror -std=c99` with zero unused neural-network weight tables).

States are stored in signed 32-bit `Q16.16` with dynamic pre-scaling ($N_{cells}/64$, $R_{LP}/32$, $S_{SRC}/8$, pH/8), `RK4` slopes in `Q8.24` ($2^{-24} \approx 5.96 \times 10^{-8}$) with 64-bit ΣΔ remainder tracking, and the 8×8 EKF covariance matrix $\mathbf{P}$ in signed `Q12.20` (spanning $[10^{-6}, 2048)$ so both initial R_{LP} variance 4.0 and optical `ORR` noise variance 1.44×10^{-4} are resolved without overflow or underflow). Sequential scalar measurement updates use an exact 64-bit integer square root (`isqrt64_u32`) for Huber innovation clipping ($|\nu_i| \leq 2.5\sqrt{S_i}$) and an exact division-free integer biofouling check ($2\nu_1^2 > 27S_1 \iff \text{NIS}_1 > 13.5$). Table 5 summarizes exact metrics emitted by `cart_bioreactor_hil_testbench.c --json`:

1. **Zero-Heap Static SRAM Footprint:** Removing unused neural projection weights shrinks `sizeof(coralnpu_cart_dtcm_bank0_workspace_t)` to 2,704 B (33.01% of an 8,192 B SRAM bank), encompassing all `Q16.16` state/checkpoint vectors, `Q12.20` 8×8 covariance and process-noise matrices, `Q8.24` `RK4`/ΣΔ registers, 64-frame persistence buffers, and stack margin.

2. **Deadband-Free `RK4` Parity Across Step Sizes:** At $N = 1{,}000$ steps, `Q16.16`+ΣΔ matches `FP64` switching time to the exact step (66.024 h), final expansion (62.7802× vs. 62.7979×), and T_{SCM} (73.1881% vs. 73.1989%), keeping normalized ℓ_2 trajectory error flat at 0.0078%–0.0114% across $N \in \{500, \ldots, 8{,}000\}$ (478× lower than naive `Q16.16`'s 5.2906% drift at $N = 8{,}000$).

3. **Full Integer 5-Channel Huber-EKF & NIS Handover Parity:** Across 560 closed-loop frames (168 h) on a mismatched, biofouled oncology donor ($h_0 = 51.0\%$, $\alpha_{OD} = 1.05$), the integer C99 5-channel Huber-EKF achieves $\hat{h}$ RMSE = 2.3389% vs. 2.3346% in IEEE-754 `FP64` (0.0216% **C99-vs.-`FP64` trajectory difference**), triggering $\text{NIS}_1 > 13.5$ optical handover (58.20 h) and closed-loop switching (84.00 h) at the exact same frames as `FP64`.

Table 5: **Freestanding FPU-Free C99 Fixed-Point Plant and 5-Channel Huber-EKF Verification.** Exact metrics emitted directly by `cart_bioreactor_hil_testbench.c --json`.

Firmware / Numerical Verification Metric	IEEE-754 FP64	Integer C99 (Q16.16/Q12.20/Q8.24)
Zero-Heap Static SRAM Workspace (sizeof)	–	2,704 B/8,192 B (33.01%)
Nominal Gate Switch Time ($N = 1{,}000$)	66.024 h	66.024 h
Nominal Fold Expansion N(168 h)	62.7979×	62.7802×
Nominal T_{SCM}(168 h) / h(168 h)	73.1989% / 6.7257%	73.1881% / 6.7253%
RK4 ℓ_2 Error ($N = 500$ / 1,000 / 2,000)	Ref.	0.0085% / 0.0082% / 0.0078%
RK4 ℓ_2 Error ($N = 4{,}000$ / 8,000)	Ref.	0.0114% / 0.0111%
Naive Q16.16 Error ($N = 1{,}000$ / 8,000)	Ref.	0.6412% / 5.2906% *(drift)*
5-Ch Huber-EKF $\hat{h}$ RMSE (Biofouled Run)	2.3346%	2.3389% (0.0216% diff)
NIS > 13.5 Handover / Closed-Loop Gate	58.20 h / 84.00 h	58.20 h / 84.00 h
Acidosis Fault min pH (No Flush / Flush)	6.1366 / 6.9106	6.1366 / 6.9106

7 Limitations, Postulated Ex Vivo Falsification Blueprint, and Reproducibility

7.1 Model Limitations and Modeling Assumptions

Several limitations bound the interpretation of these computational results:

1. **Absence of Wet-Lab Execution:** No primary human T cells, CAR lentiviral vectors, or bioreactor cassettes were experimentally cultured in this study. Rate constants tagged [**F**] in Table 1 represent phenomenologically calibrated couplings synthesized from separate immunological and structural reports [1, 3, 4, 7] rather than joint system-identification fits to a single time-series dataset.

2. **Deterministic Population-Mean Approximation:** Equations (11)–(18) track mean intracellular organelle mass and population-level T_{SCM}/TOX^+ fractions, neglecting clone-to-clone CAR copy-number variance, $CD4^+$/$CD8^+$ subset interactions, and asymmetric mitochondrial partitioning at cytokinesis.

3. **Optical Redox Confounding at High Suspension Density:** While cell concentration cancels in first-order two-band autofluorescence ORR (23) [14], inner-filter reabsorption at ultra-high hollow-fiber densities ($> 3 \times 10^7$ cells/mL) and riboflavin/phenol-red media fluorescence require synchronized cell-free perfusate blanking in physical hardware.

7.2 Postulated Ex Vivo Benchtop Falsification Protocol

To enable direct experimental falsification without requiring *in vivo* animal studies, our simulations generate concrete, quantitative predictions testable in a 5-arm parallel perfusion bioreactor study using primary human CD19-BBζ CAR-T cells from $n \geq 20$ pretreated oncology donor apheresis samples:

- **Arm 1 (Standard Warburg Control):** 25 mM glucose + 300 IU/mL IL-2 (0–168 h). *Prediction:* High expansion (> 70×), high $TMRE^{low}$/$MitoTracker^{high}$ depolarized fraction (> 50%), $T_{SCM} < 10\%$.

- **Arm 2 (Continuous Urolithin A Monotherapy):** Media-A (25 μM urolithin A) continuously (0–168 h) in 25 mM glucose. *Prediction:* Clearance of $TMRE^{low}$ mitochondria (< 5%) but incomplete T_{SCM} conversion ($\sim$ 25–35%) due to persistent glycolytic TOX^+ acquisition.

- **Arm 3 (Fixed 68 h Open-Loop Switch):** Media-A (0–68 h) switched blindly at 68.0 h to Media-B (6.5 mM glucose + 18.5 mM galactose, 1.0 μM AZD7545, 1.0 mM nicotinamide riboside, 10 ng/mL IL-7/IL-15).

- **Arm 4 (Chemical-Probe-Free ORR-EKF Closed-Loop + Adaptive 168–240 h Harvest):** Closed-loop ORR+OUR_{vol}+OD+pH Huber-EKF switching (24) from Media-A to STAC-free Media-B (AZD7545 + NR only, avoiding SRT1720 off-target risk [12]), with OD-triggered harvest up to 240 h for slow-dividing donors. *Falsification Criterion:* The model is falsified if Arm 4 fails to increase the 5-criteria pass rate ($N \geq 50\times$, $CD45RA^+CD62L^+TCF7^+ \geq 68\%$, $TOX^+ \leq 15\%$, Seahorse SRC $\geq 3.5\times$) relative to Arm 3, or if hypoproliferative ($\mu_{glyc} < 0.85\mu_{nom}$) donors fail to reach $\geq 50\times$ viable fold expansion during the 168–240 h consolidation window (Figure 3B).

- **Arm 5 (SIRT1-KO / Glu230Lys & Single-Target Ablation Controls):** Closed-loop switching omitting either AZD7545 or NR (testing the single-target ablation collapse predicted in Table 3), plus SIRT1 CRISPR knockout / Glu230Lys point-mutant specificity controls if synthetic STACs (4TQ/SRT2104) are evaluated [13].

7.3 Code and Reproducibility Availability (MIASE)

All structural audits, 8-state FP64 simulations, 200-variable discrete-adjoint L-BFGS-B optimizations, Observability Gramians, $N = 1{,}000$ donor Monte Carlo runs and exact paired McNemar tests (PRNG seed 20261111), publication vector figures, and freestanding C99 fixed-point verification benchmarks are deterministically regenerated from source via `simulator/generate_paper2_verified_benchmarks.py` and `build/coralnpu/cart_bioreactor_hil_testbench.c` at https://github.com/deesatzed/mitorestorer.

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Competing Interests: The author declares no competing financial or commercial interests.

Ethics Statement: This study consists exclusively of theoretical modeling, structural bioinformatics on public Protein Data Bank / AlphaFold entries, numerical ODE simulation, and host-emulated firmware verification; no human participants, primary donor tissues, clinical cohorts, or live animals were involved.

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